ISWE406P

Agile Development Process and DevOps Lab

Assessment-4

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Class No.: VL2025260502390
Slot: L3+L4

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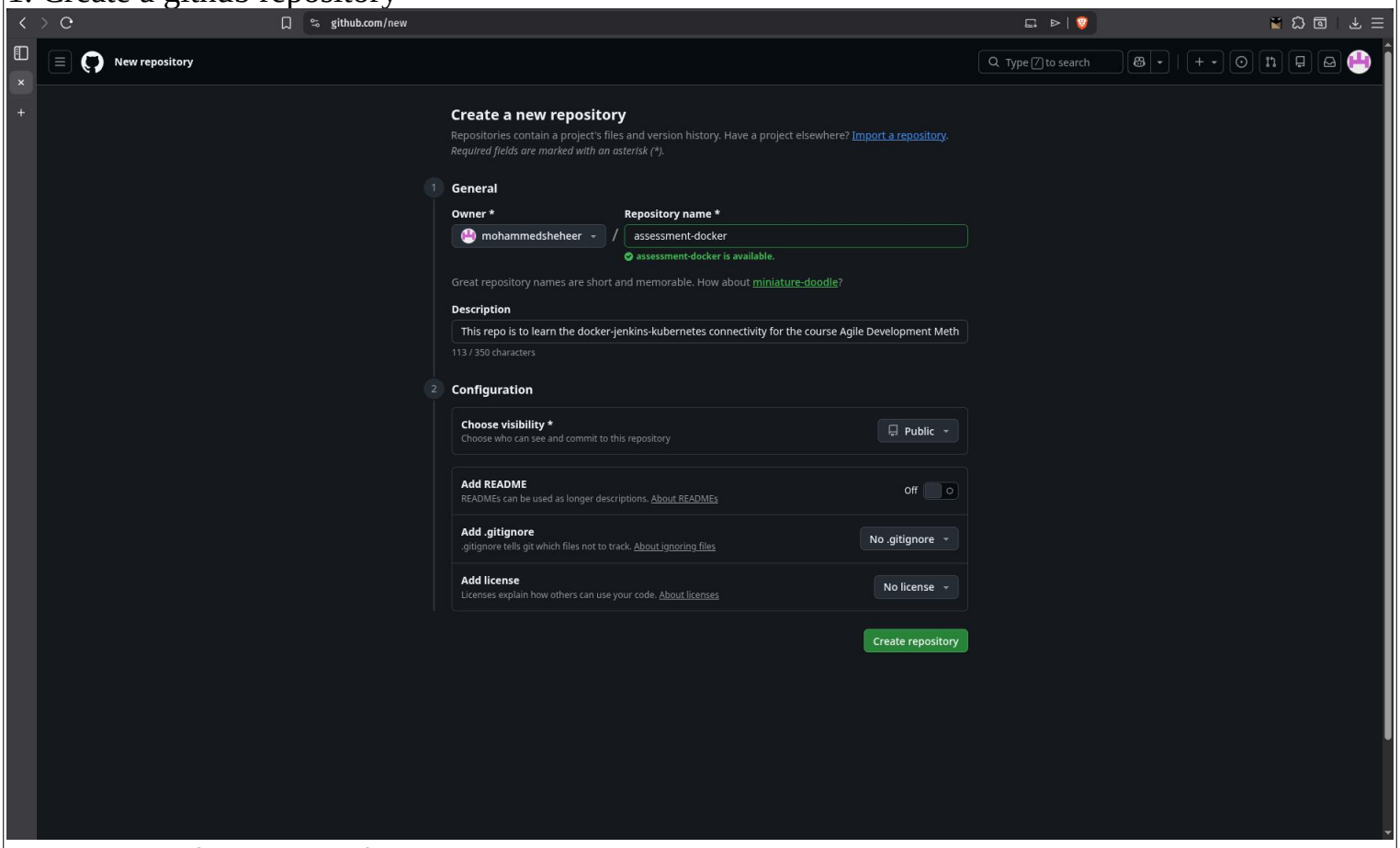
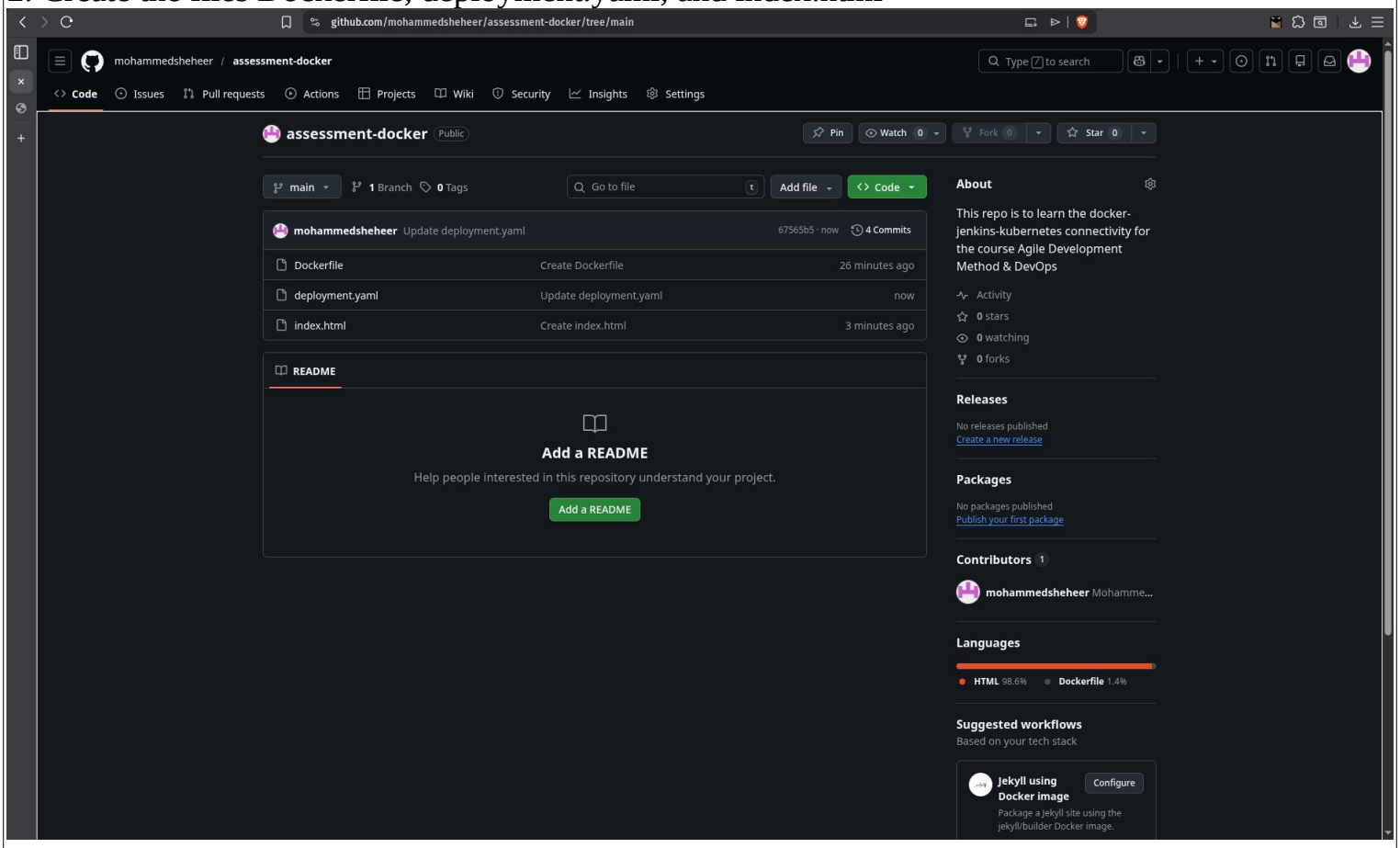
Reg No: 23MIS0170

1. Create a github repository



2. Create the files Dockerfile, deployment.yaml, and index.html



3. Create Dockerfile

github.com/mohammedsheheer/assessment-docker/blob/main/Dockerfile

mohammedsheheer / assessment-docker

Code | Issues | Pull requests | Actions | Projects | Wiki | Security | Insights | Settings

main | assessment-docker / Dockerfile

mohammedsheheer Create Dockerfile — 09485bb · 27 minutes ago — History

Code | Blame — 3 lines (3 loc) · 77 Bytes

```
FROM nginx:latest
COPY index.html /usr/share/nginx/html/index.html
EXPOSE 80
```

4. Create deployment.yaml

github.com/mohammedsheheer/assessment-docker/blob/main/deployment.yaml

mohammedsheheer / assessment-docker

Code | Issues | Pull requests | Actions | Projects | Wiki | Security | Insights | Settings

main | assessment-docker / deployment.yaml

mohammedsheheer Update deployment.yaml — 7394d5f · 29 minutes ago — History

Code | Blame — 32 lines (32 loc) · 558 Bytes

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: html-demo
spec:
  replicas: 2
  selector:
    matchLabels:
      app: html-demo
  template:
    metadata:
      labels:
        app: html-demo
    spec:
      containers:
        - name: html-demo
          image: mohammedsheheer23mis0170/assessment-docker:latest
          ports:
            - containerPort: 80
---
apiVersion: v1
kind: Service
metadata:
  name: html-demo-service
spec:
  type: NodePort
  selector:
    app: html-demo
  ports:
    - port: 80
      targetPort: 80
      nodePort: 30010
```

5. Create index.html

github.com/mohammedsheheer/assessment-docker/blob/main/index.html

mohammedsheheer / assessment-docker

Code | Issues | Pull requests | Actions | Projects | Wiki | Security | Insights | Settings

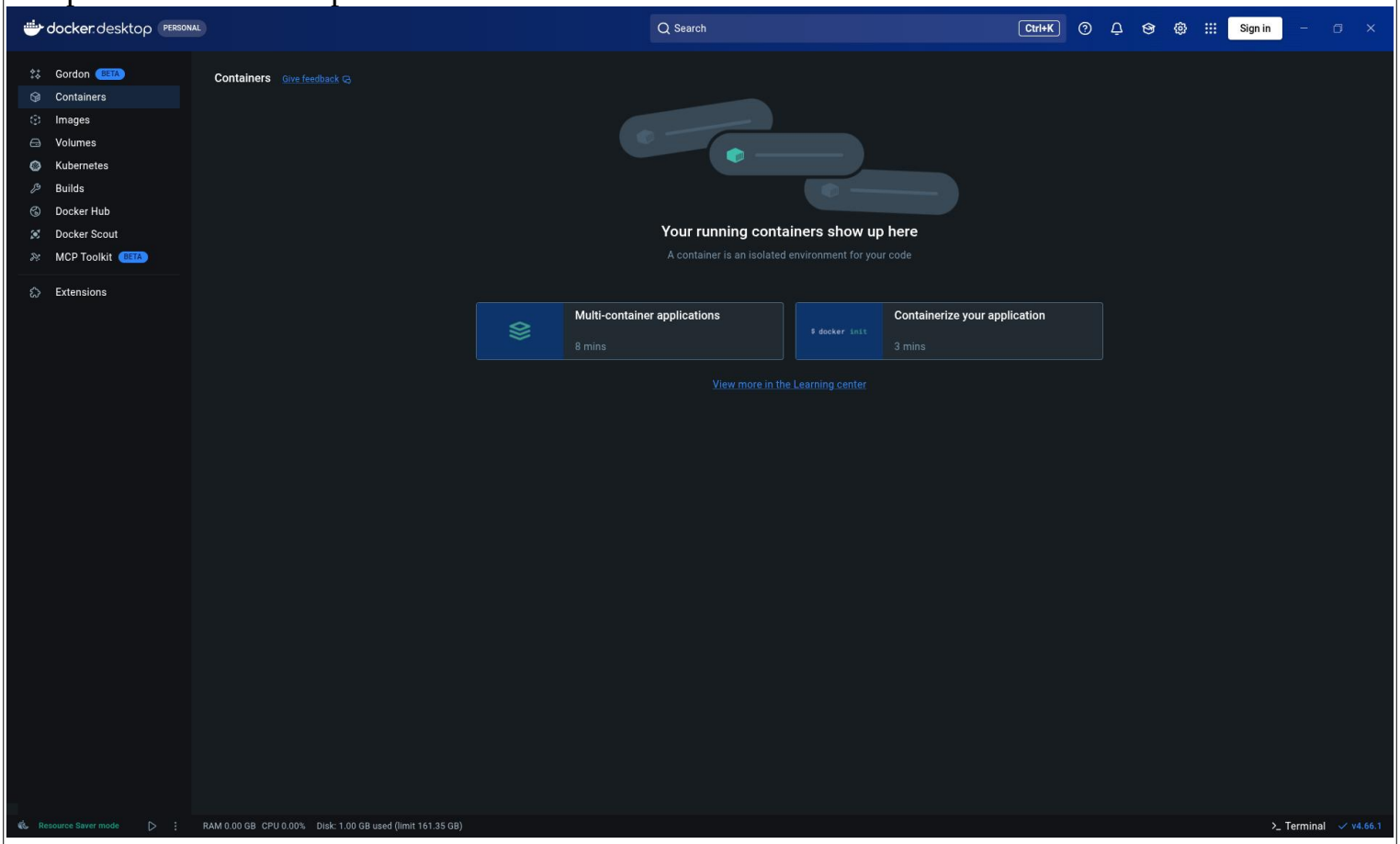
main | assessment-docker / index.html

mohammedsheheer Create index.html — ee0dfaa · 5 minutes ago — History

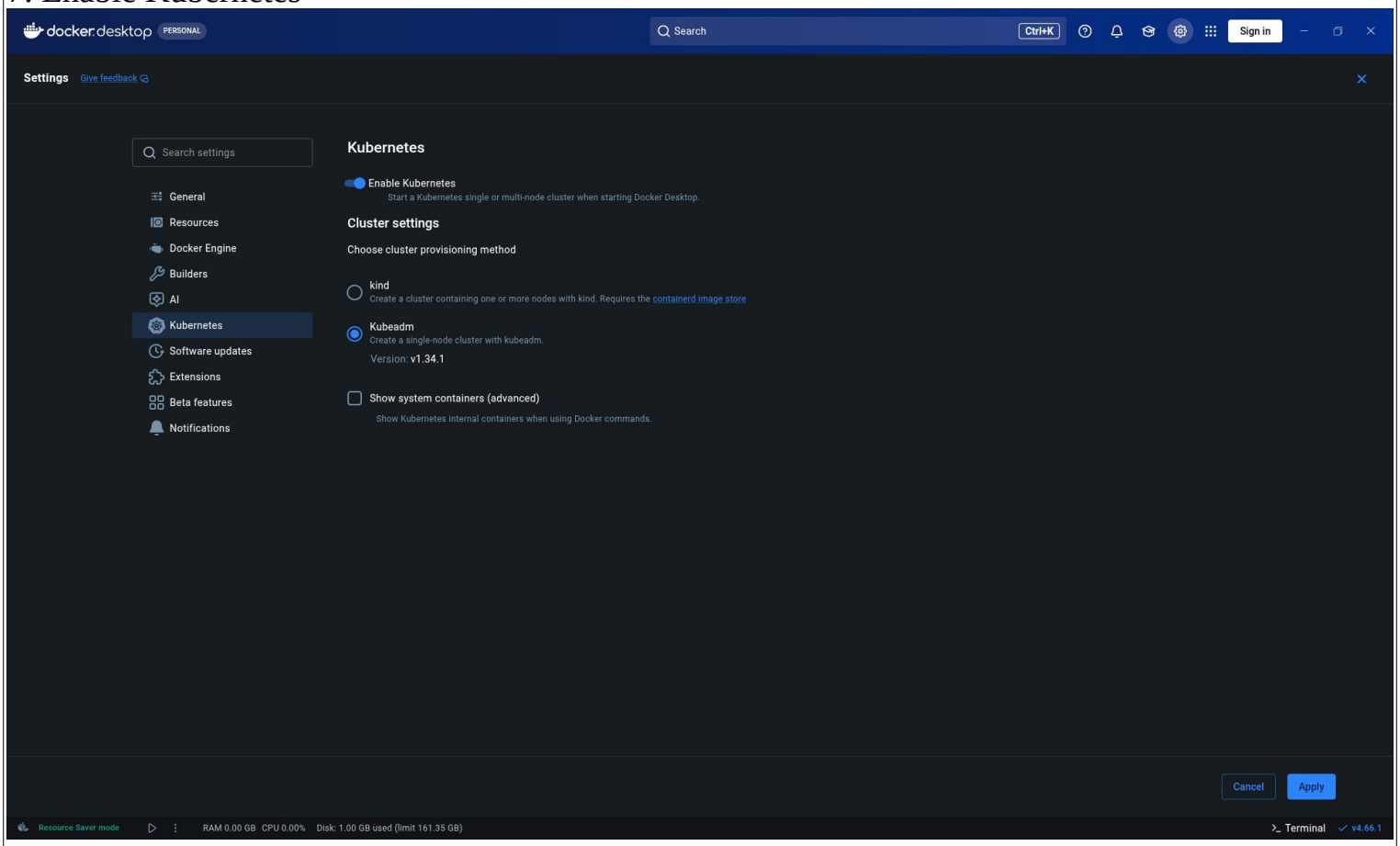
Code | Blame — 186 lines (163 loc) · 5.15 KB — Raw

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Mohammed Sheheer | Software Engineer</title>
    <style>
        :root {
            --bg: #0f172a;
            --card-bg: #1e293b;
            --accent: #facc15;
            --text-main: #f8fafc;
            --text-dim: #94a3b8;
        }

        body {
            background-color: var(--bg);
            color: var(--text-main);
            font-family: 'Segoe UI', Roboto, Helvetica, Arial, sans-serif;
            line-height: 1.6;
            margin: 0;
            display: flex;
            justify-content: center;
            padding: 40px 20px;
        }

        .container {
            max-width: 800px;
            width: 100%;
        }

        header {
            border-bottom: 2px solid var(--card-bg);
            padding-bottom: 20px;
            margin-bottom: 30px;
        }

        h1 {
            color: var(--accent);
            font-size: 2.5rem;
            margin: 0;
```

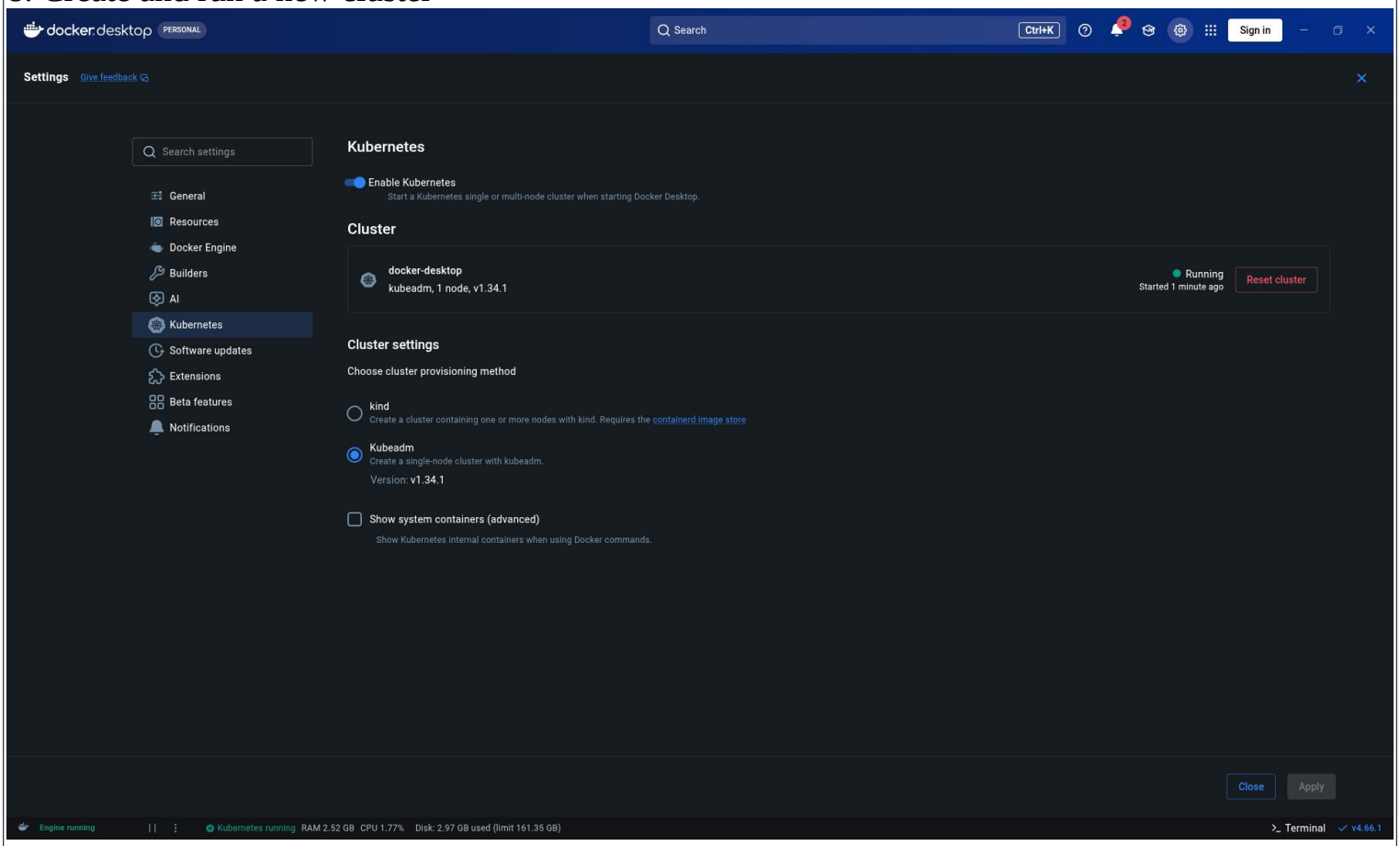
6. Open Docker Desktop



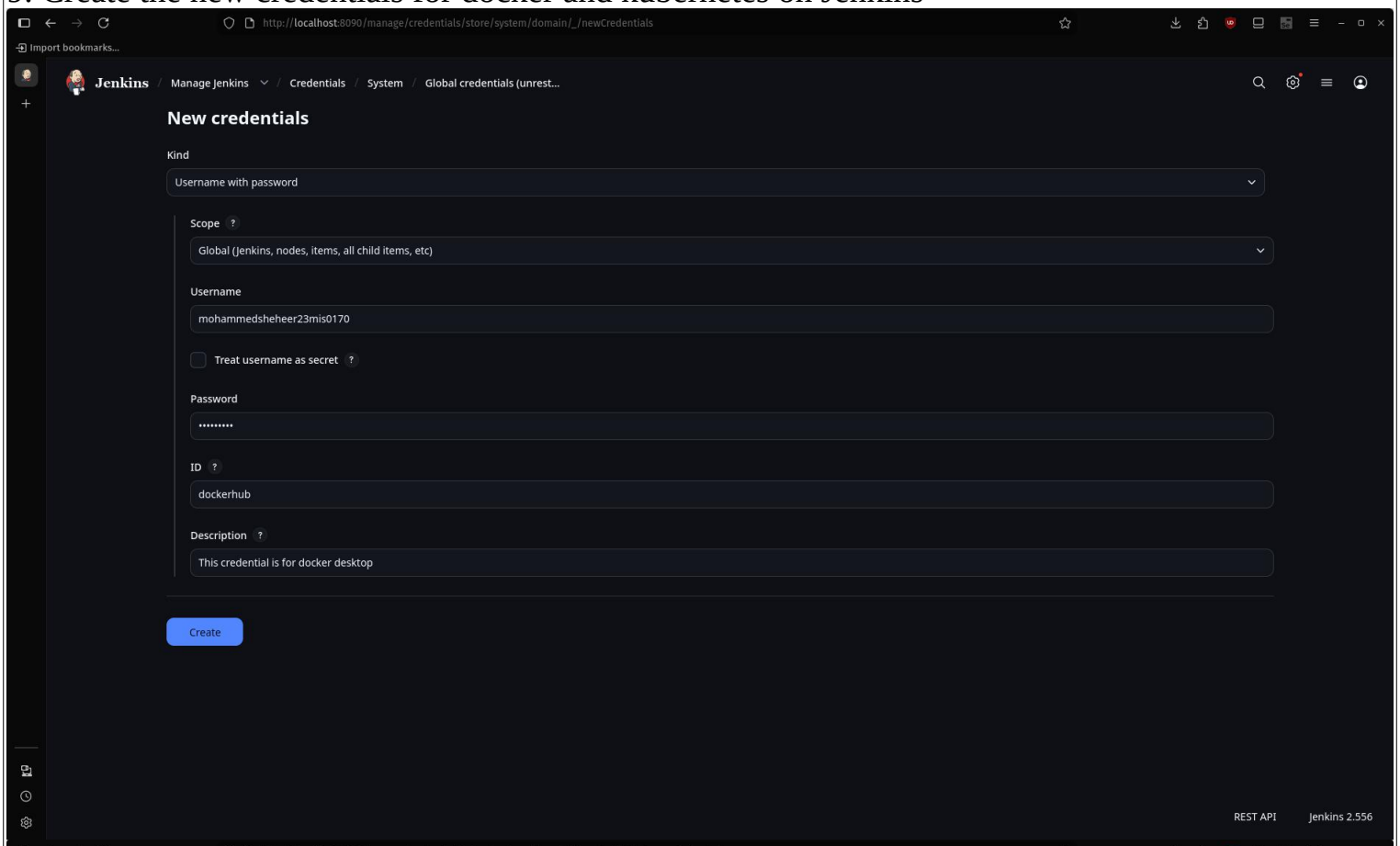
7. Enable Kubernetes

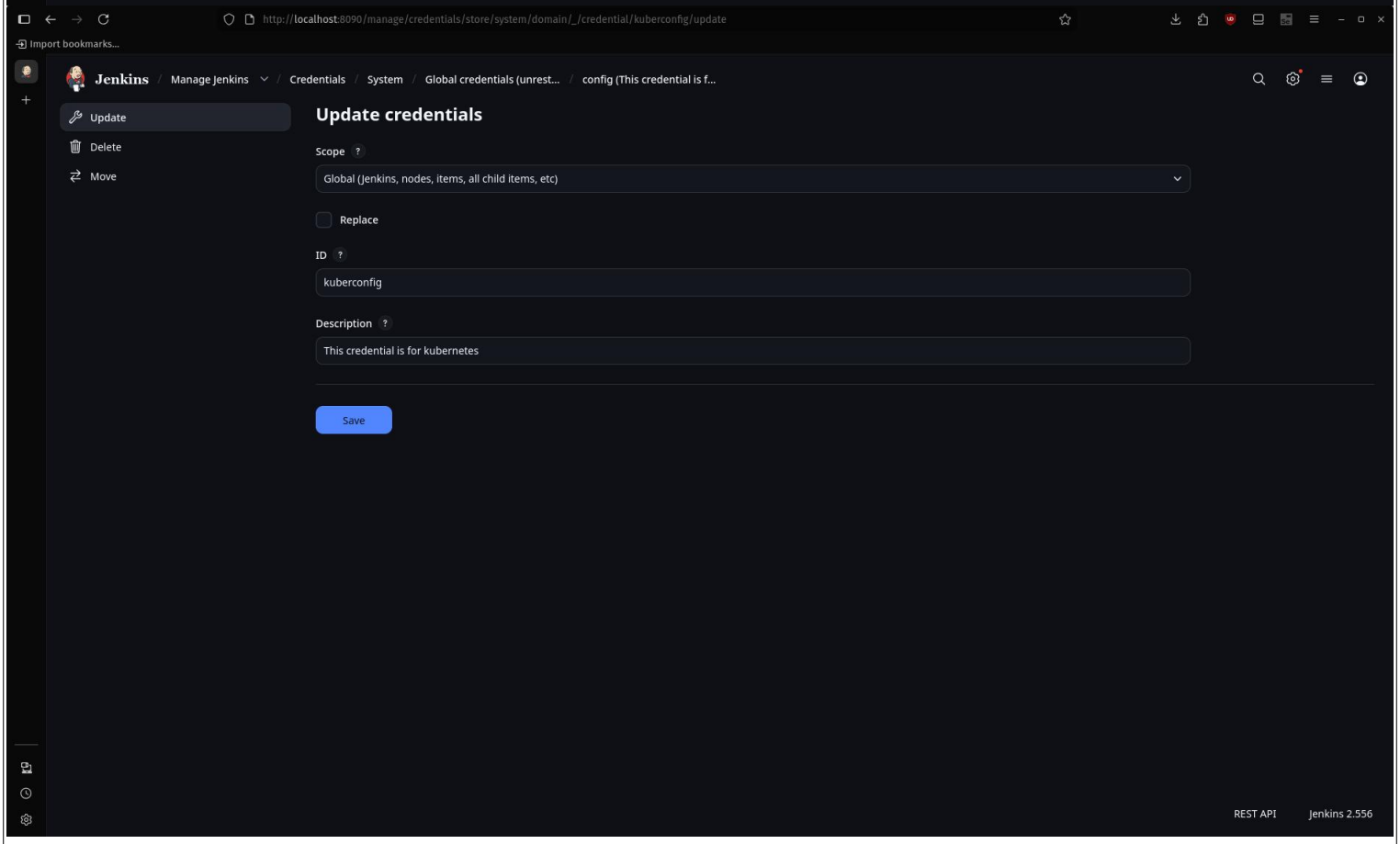


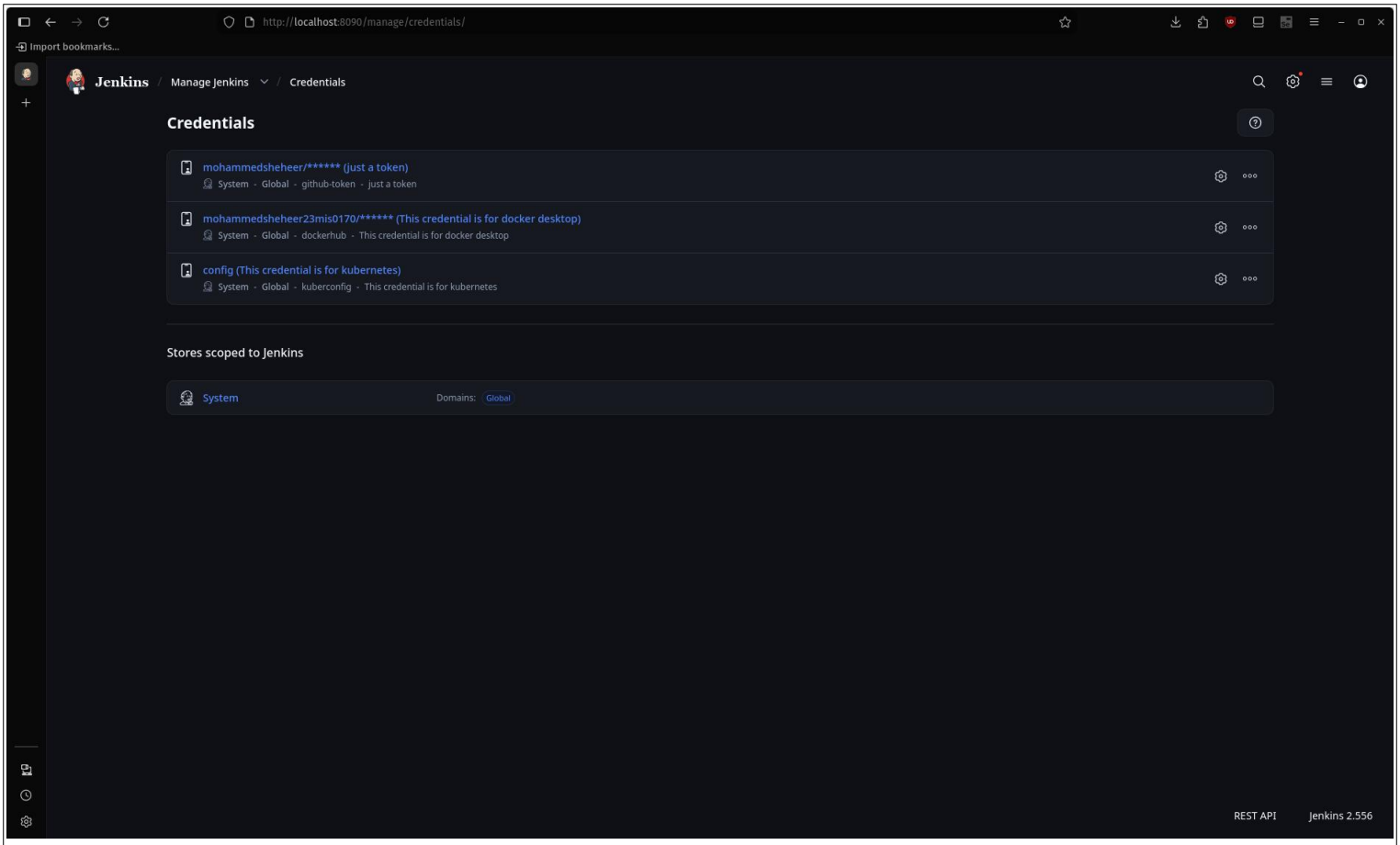
8. Create and run a new cluster



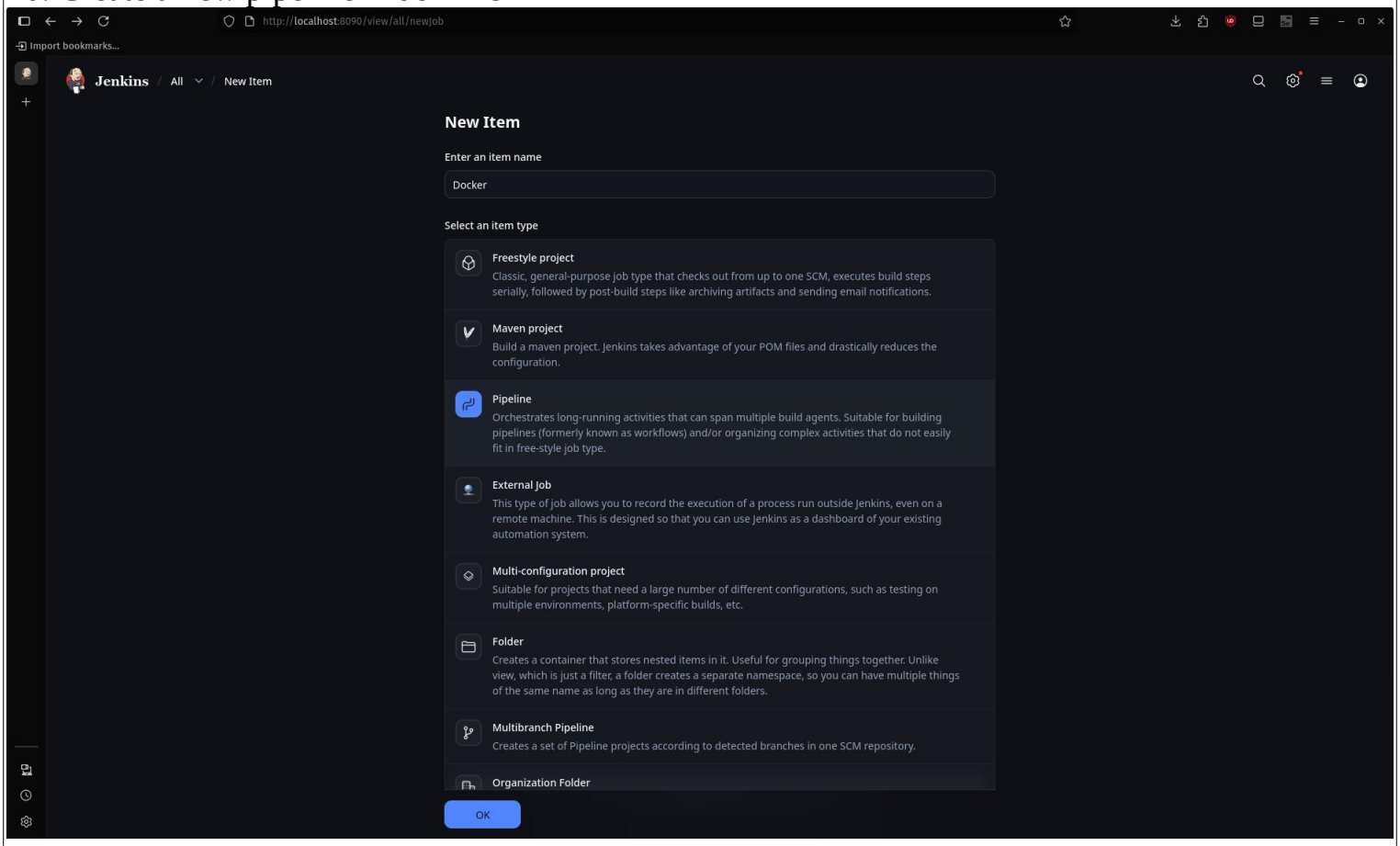
9. Create the new credentials for docker and kubernetes on Jenkins







10. Create a new pipeline in Jenkins



http://localhost:8090/job/Docker/configure

Import bookmarks...

Jenkins / Docker / Configure

Configure

- General
- Triggers
- Pipeline
- Advanced

General

Enabled

Description

This is a pipeline to showcase how well the jenkins - Docker - Github - Kubernetes connectivity helps us developers in automating daily tasks.

Plain text Preview

- [ ] Discard old builds ?
- [ ] Do not allow concurrent builds
- [ ] Do not allow the pipeline to resume if the controller restarts
- [x] GitHub project

Project url ?

https://github.com/mohammedsheheer/assessment-docker.git

Advanced

- [ ] Pipeline speed/durability override ?
- [ ] Preserve stashes from completed builds ?
- [ ] This project is parameterized ?
- [ ] Throttle builds ?

Triggers

Set up automated actions that start your build based on specific events, like code changes or scheduled times.

- [ ] Build after other projects are built ?

Save Apply

http://localhost:8090/job/Docker/configure

Import bookmarks...

Jenkins / Docker / Configure

Configure

- General
- Triggers
- Pipeline
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Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script

Script ?

try sample Pipeline...

```
                    bat 'docker push %DOCKER_IMAGE%' [cite: 169]
                }
            }
        }

        stage('Deploy to Kubernetes') {
            steps {
                withCredentials([file(credentialsId: 'kuberconfig', variable: 'KUBECONFIG')]) { [cite: 175, 176]
                    bat 'set KUBECONFIG=%KUBECONFIG%' [cite: 178]
                    bat 'kubectl apply -f deployment.yaml --validate=false' [cite: 179]
                }
            }
        }
    }
}
```

- [x] Use Groovy Sandbox ?

Pipeline Syntax

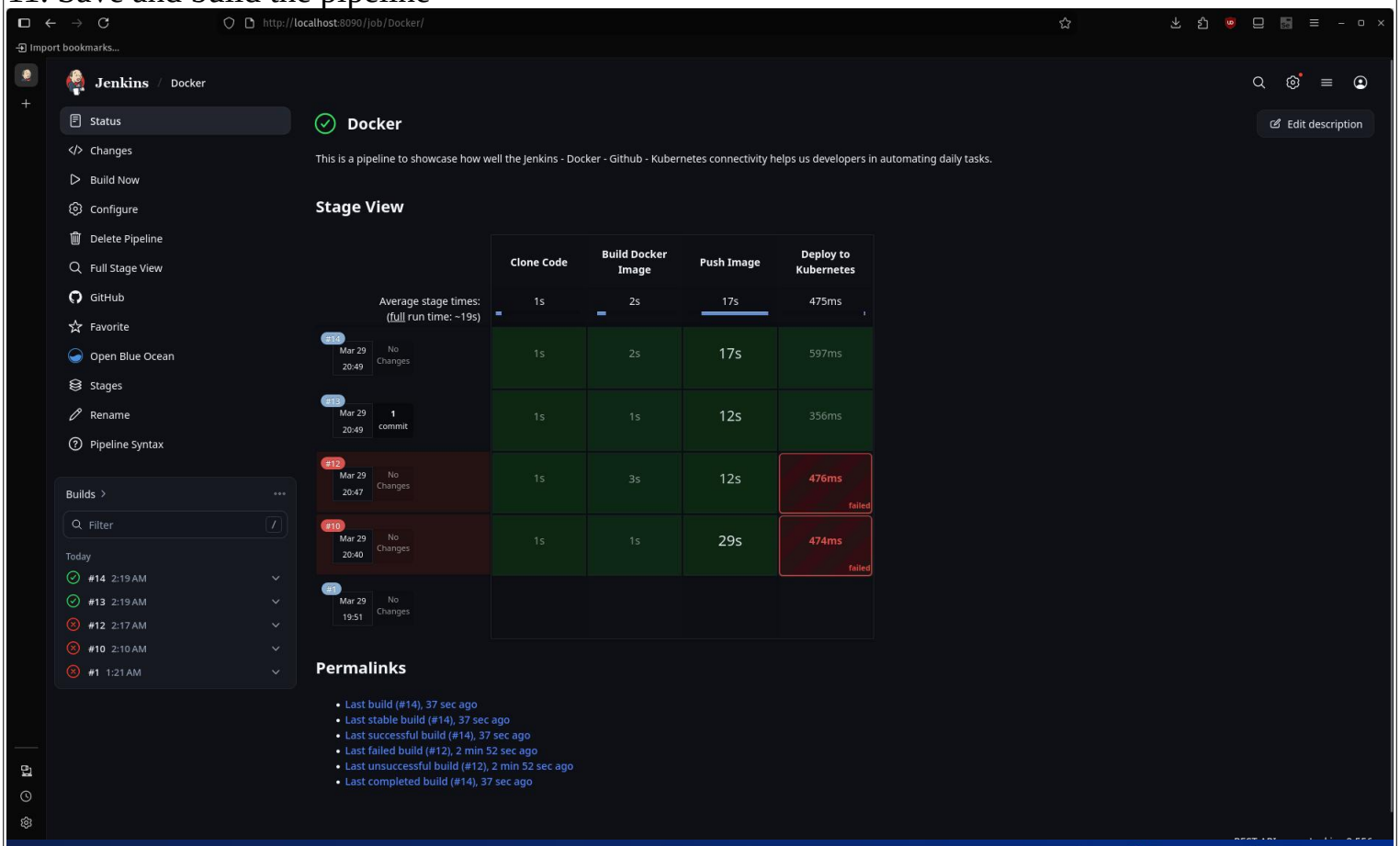
Advanced

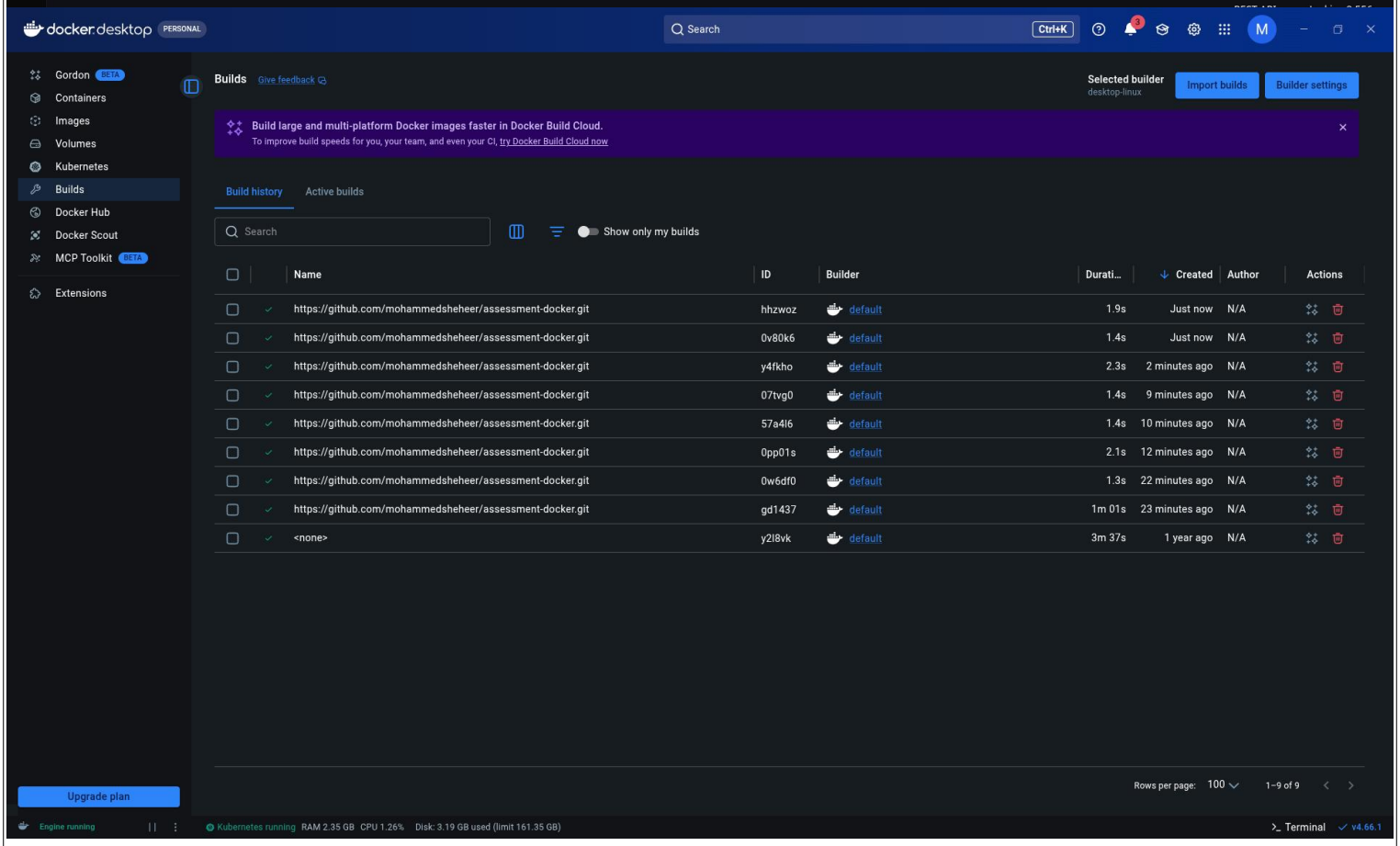
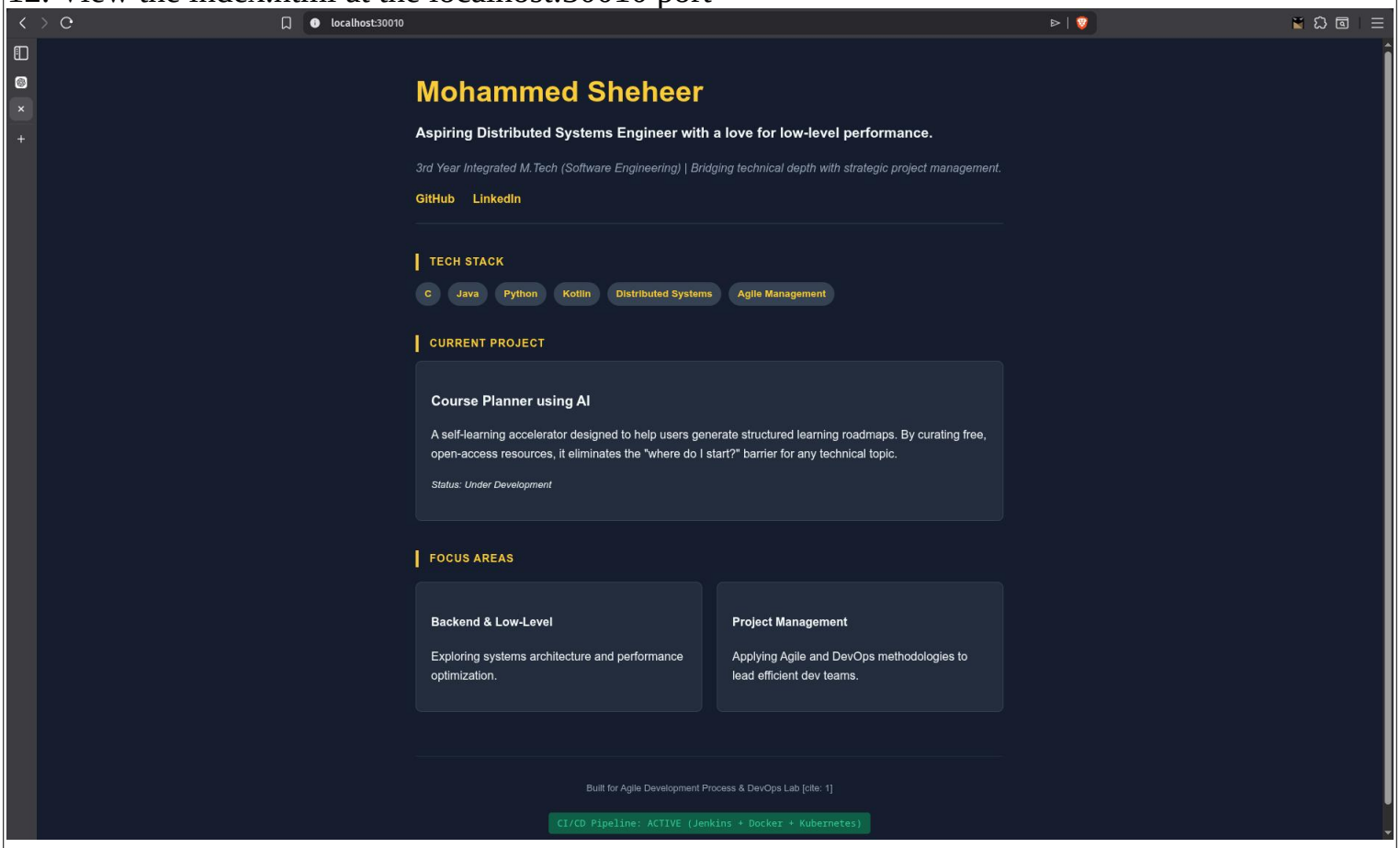
Advanced

Save Apply

REST API Jenkins 2.556

11. Save and build the pipeline





12. View the index.html at the localhost:30010 port